

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

Jay walks at a speed of **3** miles per hour and runs at a speed of **5** miles per hour. He walks for *w* hours and runs for *r* hours for a combined total of **14** miles. Which equation represents this situation?

Answer

- A. $3w + 5r = 14$
- B. $\frac{1}{3}w + \frac{1}{5}r = 14$
- C. $\frac{1}{3}w + \frac{1}{5}r = 112$
- D. $3w + 5r = 112$

Correct Answer: A

Rationale

Choice A is correct. Since Jay walks at a speed of **3** miles per hour for *w* hours, Jay walks a total of **3w** miles. Since Jay runs at a speed of **5** miles per hour for *r* hours, Jay runs a total of **5r** miles. Therefore, the total number of miles Jay travels can be represented by **3w + 5r**. Since the combined total number of miles is **14**, the equation **3w + 5r = 14** represents this situation.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.